Abstraction vs. Computation in Mathematics Education: Evidence from New Math

Colin Adams*

Working Paper†

Abstract

I study the tradeoff between abstract reasoning and computation in mathematics education using the largest change in America's mathematics curriculum, the New Math movement of the 1960s. New Math shifted the K-12 math curriculum away from arithmetic toward abstract concepts such as set theory and formal logic. I exploit New Math's quick rollout, prompted by the Soviets' launch of *Sputnik 1*, and subsequent withdrawal to estimate abstraction's effect on the set of tasks of individuals in their occupation. I find exposure to an abstract math curriculum, via New Math, to cause individuals to sort into occupations with less calculation, with no affect on sorting into analytic (i.e. abstract math) jobs. This also serves as the first to empirically analyze the New Math movement and offers evidence on a curriculum tradeoff facing modern educators.

JEL: I21, I28, J24, N32

*Ph.D. Student in Economics, Florida State University (colinpadams.com | ca23a@fsu.edu) I am grateful to Dr. Mark Isaac, Dr. Shawn Kantor, Dr. Carl Kitchens, Dr. Christine Mokher, and Dr. Luke Rodgers for their invaluable advice and support on this project.

†Last updated on March 18, 2026. You can see the most recent version of this paper here.

"The Soviets have put a satellite into space. It does not raise the specter of immediate danger. But it does raise deep concern—about our scientific and technical progress, our educational system, and our ability to remain the world's leading nation."

— Dwight D. Eisenhower, *Statement Upon Signing the National Defense Education Act*, 1958

I Introduction

There exists a vast literature on the effects of inputs into the education production function. Inputs such as teachers (Rivkin, Hanushek, and Kain 2005; Hanushek and Rivkin 2010; Chetty, Friedman, and Rockoff 2014), facilities (Hanushek 2003; Baron 2022), and class size (Angrist and Lavy 1999; Hoxby 2000; Rivkin, Hanushek, and Kain 2005) have all been found to affect student outcomes. Effects of the education production function itself, through curricula and pedagogy, remain unclear. Haeck, Lefebvre, and Merrigan (2014) find that replacing traditional instruction with student-led, problem-based learning worsened math scores and Hahm (2026) finds adoption of the Common Core to increase annual wages and increase communication.

A separate literature connects education to labor market outcomes through occupational tasks. Autor, Levy, and Murnane (2003) defined jobs as a bundle of various tasks which can be measured. Work by David Autor, David Dorn, David Deming, and others have shown improvements in technology to substitute for routine work and complement social and abstract work (e.g., Autor and Dorn 2013; Autor and Handel 2013; Deming 2017). Goldin and Katz (2008) combine these literatures, showing that educational attainment determines long-run wages and the task composition of the labor force.

This paper contributes to the education and task literature by asking whether the curriculum itself can affect task composition through occupational sorting, using plausibly exogenous variation in curriculum content. I exploit the largest and best funded reform to mathematics education in the U.S. being the New Math movement of the 1960s (Raimi 1996). New Math represented a temporary change in mathematics curriculum in America following the Soviet Union's launch of *Sputnik 1* in 1957. *Sputnik* intensified fear that America was falling behind the Soviets. Prompted by fear that the US was falling behind technologically, the federal government passed the National Defense Education Act providing generous funding to US education reforms, especially in math and science. At the center of mathematics reform was New Math whose rapid implementation provides an ideal setting for curriculum evaluation.

New Math was a change in curriculum from rote calculation and memorization to abstract concepts such as set theory, different number systems, and proofs for students as young as kindergarten. The curriculum shifted to teach third and fourth year college math classes by the time students graduated high school in hopes of shifting the set of skills available in the US labor market towards more mathematicians and scientists. While New Math programs received generous funding following *Sputnik*, backlash from mathematicians, teachers, and parents led to it being almost non-existent by the mid 1970s.

This paper investigates the shift in tasks completed by individuals exposed to New Math in elementary and secondary school. I exploit the exogenous timing of New Math's implementation, spurred by *Sputnik*, and use historical documents from prominent New Math groups and state education departments in Ohio to measure individuals' exposure to abstract mathematics instruction. Following the task literature, I estimate New Math's effects on the composition of tasks individuals later perform in their occupations. I find that New Math exposure reduces the likelihood of sorting into calculation-heavy occupations (e.g., bookkeepers, cashiers, and clerks) with no corresponding effect on the probability of sorting into abstract math-oriented occupations (e.g., mathematicians, physicists, or economists). These results, consistent with critics who argued New Math damaged computational skills without delivering gains in abstract reasoning, provide new evidence that curriculum and pedagogy shape occupational sorting and constitute the first empirical work on the effects of the New Math movement.

II New Math Curriculum

The introduction of New Math shifted the way mathematics was taught in K-12 from using math as a way of calculation to math as a logical system, more similar to how math is taught in universities. New Math curricula differed from this in three fundamental ways. First, New Math prioritized abstract mathematical structures such as set theory, symbolic logic, and alternative number bases over computation, algebraic manipulation, and Euclidean geometry (Kline 1973). New Math treated abstraction as a way to learn calculation rather than as advanced topics reserved for college courses.

Second, where traditional curricula aimed for computational proficiency through memorization and drills, New Math emphasized deductive reasoning to teach students why mathematical procedures work rather than how to execute them (Miller 1990). The third difference was in mathematical language and formalism. New Math introduced formal mathematical notation and terminology, often centered in set theory, in elementary school. Rather than teaching addition through counting, New Math taught addition through the union of disjoint sets, shown in Figure 1 for fourth graders. Another example is how the Pythagorean Theorem was taught using geometric decomposition rather than through a formula. This was done using Figure 2 which includes a rhombus inside of a square where angles y and x form a right angle. The area of the large square equals that of the the small square plus the four triangles in gray resulting in equation 1 which is the Pythagorean Theorem when simplified. Other examples of New Math teachings are in the curriculum

Figure 1: New Math Addition

A binary operation is *commutative* if changing the order of the numbers operated on does not affect the result. Since the sum of two numbers is the cardinal number of the union of two disjoint sets, we can show that addition of whole numbers is a commutative operation. First we note that $A \cup B$ is the same set as $B \cup A$, for any two sets A and B, because however we join A and B, we still have the same elements in the union. Second, suppose that a and b are two whole numbers, and that A and B are two disjoint sets whose cardinal numbers are a and b, respectively. Then by definition of addition:

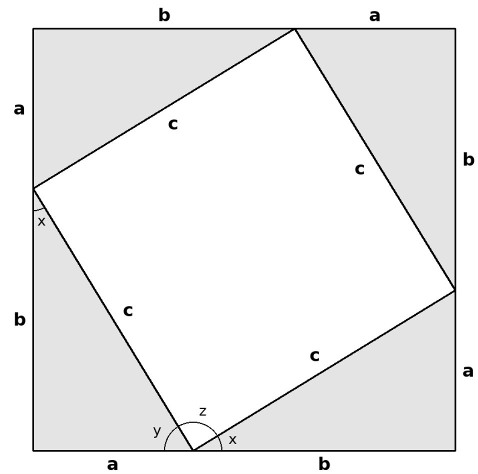
$a + b$ is the cardinal number of $A \cup B$.
$b + a$ is the cardinal number of $B \cup A$.

These two cardinal numbers are the same, since $A \cup B = B \cup A$. Thus $a + b$ is equal to $b + a$, no matter what whole numbers a and b are. This is what is meant by the statement, "Addition of whole numbers is commutative." This observation is important because, as we will see later, not all operations are commutative.

Note: Source: Greater Cleveland Mathematics Program 1962.

appendix.

Figure 2: New Math's Pythagorean Theorem Figure



$$(a+b)^2 = a^2 + 2ab + b^2 = c^2 + 4\left(\frac{1}{2}\right)ab \tag{1}$$

Note: Source: Phillips 2014.

III New Math Exposure

III.A Beginnings

Beginning in the 1940s, many mathematicians affiliated with the American Mathematical Society (AMS) and the Mathematical Association of America (MAA) began voicing dissatisfaction with K–12 mathematics education in the United States. They believed school curricula focused excessively on rote arithmetic and calculation, neglecting newer mathematical developments such as set theory and abstract algebra.[1] In response, Max Beberman founded the University of Illinois Committee on School Mathematics (UICSM) in 1951. UICSM sought to develop a new pedagogy for high school mathematics and produced a wide array of instructional materials (UICSM Mathematics Project 1970). Beberman and UICSM designed the first curriculum of what would later become the New Math movement, also called the modern mathematics movement, emphasizing logic and number systems over computation (Phillips 2014).

A few years after UICSM's founding, the Soviet Union launched *Sputnik 1*, the world's first artificial satellite, on October 4, 1957. The launch sparked widespread fear in the United States, marking the beginning of the Space Race and leading to the creation of the National Aeronautics and Space Administration (NASA) the following year. This science oriented arms race affected all levels of American society, especially the education system, and as part of its response to *Sputnik* the federal government dramatically increased

[1]This is well documented in both peer-reviewed journals and books (e.g., Stanic and Kilpatrick 1992, Graham 2006, and Phillips 2014).

investment in science and mathematics education through the National Defense Education Act in 1958 and increased funding to the National Science Foundation (NSF). Investment from the federal government led to widespread changes among K-12 education. For example, the rate of mathematics instructional innovation in New York public schools nearly tripled in the fifteen months following *Sputnik*'s launch (Brickell 1961).

III.B Sputnik, the NSF, and the School Mathematics Study Group

The NDEA aimed to help America catch up to the Soviet Union in science and technology as the NSF's appropriation rose from $40 million in 1958 to $134 million the following year (Hunt 2025). Among the major recipients of NSF funding was Edward Begle's School Mathematics Study Group (SMSG), launched at Yale University in 1958 (Begle 1963).[2] Inspired by Beberman's earlier efforts, Begle's SMSG collaborated with mathematicians and educators to produce K–12 textbooks that emphasized set theory, non-decimal number bases, and formal logic which states began incorporating into their official curricula (Phillips 2014). While initial efforts focused on high school, the approach eventually spread across all K–12 grades. By the mid-1960s, more than 85% of U.S. public school districts used New Math instructional materials at some level (Miller 1990).

III.C Greater Cleveland Mathematics Program and Other New Math Groups

While SMSG was the largest New Math group, many others were created following *Sputnik* to revamp math education across the US. The most influential in Ohio was the Greater Cleveland Mathematics Program (GCMP) created by the Educational Research Council of Greater Cleveland (ERC), an independent, nonprofit organization, founded in 1959 by Dr. George H. Baird. GCMP was ERC's first and largest program and sought to improve the quality of elementary and secondary math education in and around Cleveland, Ohio (ERC 1963).[3]

GCMP was a self-described New Math program emphasizing reasoning with its authors including mathematicians, math educators, and other experienced teachers.[4] The group was heavily inspired by SMSG and took recommendations from the National Council of Teachers of Mathematics and the MAA (ERC 1963). Different from SMSG, the GCMP focused on elementary math with its initial rollout of K-3 programs in January 1960 and expanded to K-6 in the 1961-1962 school year. Within two years of its creation, GCMP was taught to over two-hundred thousand students around Ohio (Baird 1961).

[2] *Time* magazine labeled Edward Begle, and other leaders of the New Math movement, as "*Sputnik*-stirred mathematicians" (Time 1961).

[3] The Educational Research Council of Greater Cleveland was later renamed to the Educational Research Council of America in 1967. For this reason, I refer to the group as the Education Research Council or ERC from this point on.

[4] A list of the guidelines and principles of the GCMP are listed in Table 3 and Table 4, respectively, in the appendix.

Similar to SMSG and GCMP, two other groups' New Math curricula were taught in Ohio, although at far fewer schools. These other groups are the Teachers of Elementary Mathematics Advisory Committee (TEMAC) and the Minnesota Mathematics and Science Teaching Project (Minnemast). Each of these groups were similar to GCMP being based off of SMSG and UICSM's programs, but focused on the elementary-school curriculum. However, both TEMAC and Minnemast consisted mostly of experienced elementary math teachers, unlike other New Math groups whose writers were majority mathematicians.

III.D Back to Basics Movement

Criticism of New Math emerged quickly.[5] Detractors argued that New Math was too abstract, introduced advanced concepts before students had mastered basic arithmetic, and was often incomprehensible to both students and their parents. The largest critic was Morris Kline, chairman of NYU's mathematics department, whose book *Why Johnny Can't Add* argued that New Math was hopelessly abstract and impractical. He had many other showings against New Math, but none were bigger than his 1962 *American Mathematical Monthly* letter signed by sixty-four other mathematicians proclaiming their distain for New Math (Kline et al. 1962 and Phillips 2014). Nobel physicist Richard Feynman similarly contended that New Math was written by and for mathematicians, not children (Feynman 1965). Critics also pointed out that most teachers lacked sufficient training to effectively teach the new curriculum, compounding confusion in classrooms (Stanic and Kilpatrick 1992).

These concerns resulted in the "Back-to-Basics" movement of the early 1970s, which advocated for a return to traditional arithmetic and skills-based instruction (Offner 1982).[6] The momentum of this backlash accelerated the withdrawal of New Math from public schools, a process that had already begun in the late 1960s. New Math groups continually had their funding cut with SMSG ending in 1977. Although the ERC continued until 1984 with other programs, no GCMP documents were made after 1968.

IV Data

IV.A New Math Implementation

I measure New Math exposure using two sources of historical documents. The first is the 1966 *Catalog of Educational Changes in Ohio Public Schools*, led by Daniel Stufflebeam and commissioned by a collective

[5]Tom Lehrer, a famous satirical signer-songwriter and mathematician, released his song "New Math" in 1965 poking fun at the New Math Curriculum. In it he claims, "(New Math)'s so simple, so very simple. That only a child can do it!" Around the same time as "New Math", Peanuts, came out with seven daily comic strips emphasizing how New Math was written for mathematicians, not school children. These comic strips are included in the media appendix.

[6]Mueller (1966) claimed New Math's honeymoon to be over.

of Ohio education groups. This catalog documents all innovative changes to Ohio public schools from 1958 to 1964, based on the Ohio Educational Innovations Survey administered to all public school districts in 1964. The report categorizes changes as (1) instructional; (2) administrative; (3) pupil personnel services; (4) physical plant; (5) staff; (6) school-community relations; and (7) research.

Of Ohio's 611 school districts at the time, 452 districts (74%) responded to the survey. Stufflebeam's team rated all reported programs on innovativeness and included only those deemed innovative (rated at least three out of six) in the final catalog. This resulted in 241 districts appearing in the report. The catalog does not identify which districts responded but did not report innovations, so I classify all other districts not appearing in the report as non-adopters of New Math. District boundaries are match to shapefiles from the National Center for Education Statistics' (NCES) Common Core of Data (CCD) for 1995.[7]

From the catalog's instructional program changes, I identify districts that adopted New Math curricula. New Math had several versions developed by different curriculum groups—UICSM, SMSG, GCMP, TEMAC, and Minnemast—each explicitly mentioned in Stufflebeam's report. I classify any district explicitly mentioning these programs or using broader terms such as "new math" or "modern math" a New Math adopter. The specific grades receiving the New Math curriculum are reported for all districts in the catalog. When available, I extract implementation years, otherwise I assume implementation in 1961 which attenuates my results.[8] Beyond measuring New Math exposure, Stufflebeam's catalog enables me to control and check for potential confounders by providing comprehensive data on all other educational innovations Ohio districts adopted during this period across the seven program categories.

My second source consists of documents from the Greater Cleveland Mathematics Program (GCMP), the most popular New Math curriculum in Ohio. These documents provide cross-sections of all districts using GCMP materials from 1959 to 1968. I infer program withdrawal when a district appears in an earlier GCMP document but not in a subsequent one, indicating discontinuation prior to the later publication date. I find no documentation of New Math instruction in Ohio after 1968 and therefore assume it was not taught after that date.

All New Math implementation data is at the school district level which I aggregate up to the 1990 or 2000 Public Use Microdata Area (PUMA) level, this being my most granular residence variable in the Censes and ACSs. I do this using the 2000 Census tracts with population counts which I overlay to determine the average amount of New Math exposure among each cohort within each PUMA.[9] This assumes an even distribution of the population within each census tract. I go into more detail on this in Section V.A. Maps of Census

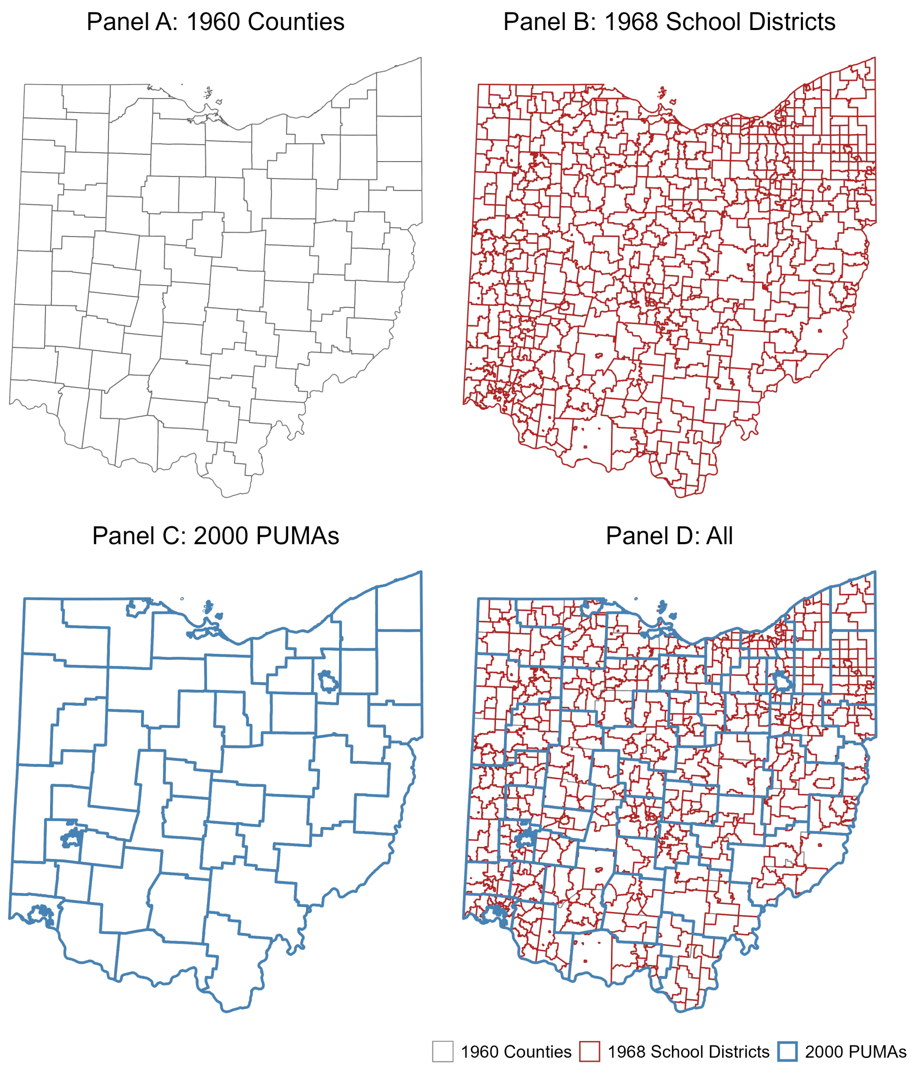
[7] I am currently working on making shapefiles of all school districts in Ohio from 1968.

[8] 1961 is the midpoint of the possible implementation period (1958 to 1964), leading to attenuation.

[9] Ideally, I would use the population distribution from 1960 Census tracts, but the 2000 Census was the first to cover Ohio completely. All previous decennial Census tracts leave significant rural areas outside any Census tracts.

tracts, school districts, and PUMAs are included in Figure 3

Figure 3: Ohio Census Tracts, School Districts, and PUMAs



Note:

IV.B Census and American Community Survey

I use the 1990 and 2000 decennial Censuses and 2005-2009 American Community Surveys (ACS) to obtain information on individuals' occupations, education, wage, and other demographics.[10] Importantly, all individuals have a Census occupation code corresponding to their current occupation. I restrict the sample to those born in Ohio to keep only individuals for whom I have knowledge of the history of New Math in Ohio while they are in school.

[10]Wages are adjusted to be in constant 2026 dollar terms and regressions only use the log of hourly wages to account for differences in work hours.

I use the 2000 Census population distribution to construct population-weighted New Math exposure within each PUMA for individuals surveyed in the 1990 and 2000 Censuses. I link Census tracts to school districts and school districts to PUMAs to determine average New Math exposure per birth cohort within each PUMA. I go into more detail on this in the following section.

All Census and ACS data is restricted to those between 24 and 54 years old, at the time of survey, following Deming (2017). This is to prevent selection effects among those remaining employed near retirement. Additionally, this prevents results from conflating New Math exposure with the convergence in task composition occurring among individuals as they reach retirement age (Autor and Dorn 2009; Nunley and Seals 2021). Additionally, I restrict to only cohorts from 1940-1962 to exploit variation only in the implementation of New Math as its withdrawal was less publicized and so I assume no New Math was taught following 1968. Thus the amount of New Math taught to cohorts starting in 1963 is less known.

IV.C Occupation Tasks and Abilities

My primary measure of occupational tasks comes from the 1998 US Department of Labor's Occupational Information Network (O*NET). The O*NET uses a random sample of US workers in each occupation to answer survey questions about their use of different abilities, skills, and knowledge in their job. Questions are rated on an ordinal scale. Following Deming (2017), the O*NET allows for the construction of ten task measures, of which I focus on four: analytic tasks, number facility, routine tasks, and social skills.[11]

Each task measure begins as the average of ordinal scales ranging from 1 to 7.[12] Measures are then converted into percentiles corresponding to their rank in the 1980 employment distribution, using the 1980 Census, to prevent ordinal measures as is common in the literature (e.g., Autor, Levy, and Murnane 2003; Autor and Dorn 2013; Deming 2017). Occupation codes, which are reported differently across biennial Censuses and ACSs, are linked using the crosswalk specified in Autor and Dorn (2013).

Other task measurements come from the 1977 Dictionary of Occupational Titles (DOT). The DOT evaluates over 12,000 occupations along 44 objective and subjective dimensions by having analysts employed by the US Employment Service conduct on-site job studies and interviews. I follow Autor, Levy, and Murnane (2003) and create five different task measures being (1) non-routine cognitive analytical (2) non-

[11]The questions corresponding to analytic tasks are (1) mathematical reasoning ability (2) mathematics knowledge and (3) mathematics skill. Routine tasks include (1) degree of automation (2) importance of repeating the same tasks and number facility measures (1) "the ability to add, subtract, multiply, or divide quickly and correctly." Social skill intensity measured by (1) social perceptiveness (2) coordination (3) persuasion and (4) negotiation.

[12]Importance of repeating tasks is on a 1 to 5 scale. Because of this all task measures are converted to be the average between 0 and 10.

routine cognitive interactive (3) routine cognitive (4) routine manual and (5) non-routine manual.[13] DOT measurements are similarly converted to percentiles using the 1980 employment distribution.

I focus on task measurements from the 1998 O*NET for two reasons. The first being that its more contemporaneous with the occupations I observe in the Census and ACS data. Secondly, O*NET task measurements more properly align with the effects on abstraction and computation to estimate New Math's effect. The analytic and number facility tasks, calculated using the 1998 O*NET, correspond to jobs with more abstract math and calculation, respectively, whereas the DOT's non-routine cognitive analytical task corresponds to the use of math more generally. Examples of jobs with high analytic and low number facility are mathematicians, physicists, economists, and lawyers, roughly aligning with jobs New Math hoped to promote. Those with low analytic and high number facility values are bookkeepers, cashiers, and clerks.

V Methodology

V.A New Math Exposure

I measure New Math exposure at the PUMA-cohort (pc) level, where PUMA boundaries correspond to either the 1990 or 2000 Census definitions, denoted by $v(t)$.[14] New Math exposure for those in a given PUMA-cohort pc and surveyed at time t, denoted by NM Exposure$_{pcv(t)}$, is shown in equation 2 and captures the expected number of grade-years a child spent under a New Math curriculum. NM Exposure$_{pcv(t)}$ is aggregated from the district-level adoption records to the PUMA level.

$$\text{NM Exposure}_{pcv(t)} = \sum_{g=0}^{12} \sum_{d} w_{dpv(t)} \times \mathbb{1}\{\text{cohort } c \text{ taught New Math in grade } g \text{ in district } d\} \tag{2}$$

NM Exposure$_{pcv(t)}$ is the sum of district-level New Math exposure within the PUMA, aggregated across all grade years (0–12). Exposure at the district level is determined from the historical records, and cohorts are assumed to have begun kindergarten (grade 0) at age five, corresponding to school year $c+5+g$. Because school district boundaries do not coincide with PUMA boundaries, district-level exposure is aggregated to the PUMA level using the population weights $w_{dpv(t)}$ defined in equation 3.

[13] Items on the 1977 DOT corresponding to each task measure are (1) general educational development - mathematical development being the level of mathematical reasoning needed to perform the job (2) direction control and planning capturing managerial and supervision (3) sets limits, tolerances, or standards requiring rule-based cognitive processing (4) finger dexterity capturing repetitive, standardized manual activity and (5) eye-hand-foot coordination capturing situational adaptability.

[14] 2005-2009 ACS data uses PUMAs from the 2000 Census. Thus, the PUMA definition year is a function of the survey year $t \in \{1990, 2000, 2005, 2006, 2007, 2008, 2009\}$.

$$w_{dpv(t)} = \frac{\mathrm{Pop}_{dv(t)} \times \dfrac{\mathrm{Area}(d \cap p_{v(t)})}{\mathrm{Area}(d)}}{\sum_{d'} \mathrm{Pop}_{d'v(t)} \times \dfrac{\mathrm{Area}(d' \cap p_{v(t)})}{\mathrm{Area}(d')}} \tag{3}$$

For district d, PUMA p, in either 1990 or 2000 Census terms ($v(t) \in \{1990, 2000\}$), the weight apportions district d's population, determined by Census tracts, to PUMA p in proportion to the share of district d's land area falling within p.[15] $\mathrm{Pop}_{dv(t)}$ is the total population of district d, constructed using Census tract populations from Census vintage v, and PUMA boundaries for a given Census year are denoted by $p_{v(t)}$. Intersecting land area is given by $\frac{\mathrm{Area}(d \cap p_{v(t)})}{\mathrm{Area}(d)}$ and is normalized so weights sum up to one across all districts overlapping PUMA p. Summing the weighted indicator across all thirteen grades yields NM $\mathrm{Exposure}_{pcv(t)}$, which is a continuous variable ranging from zero to twelve.

Because PUMA boundaries changed between the 1990 and 2000 Censuses, exposure is computed separately under each definition and matched to individuals based on their survey year. Individual i's exposure is assigned as NM $\mathrm{Exposure}_{pcv(t)}$ where p is their PUMA of residence at the time of survey t and c is their birth year. Figures 4 and 5 show the variation of NM $\mathrm{Exposure}_{pcv(t)}$ by PUMA-cohorts.

V.B Identification

I exploit the rapid rollout of New Math following *Sputnik* and the resulting variation in individuals' age at exposure to obtain quasi-random variation in New Math exposure, using Equation 4 to estimate the effect of an additional grade-year of exposure on the occupational task composition of individuals.

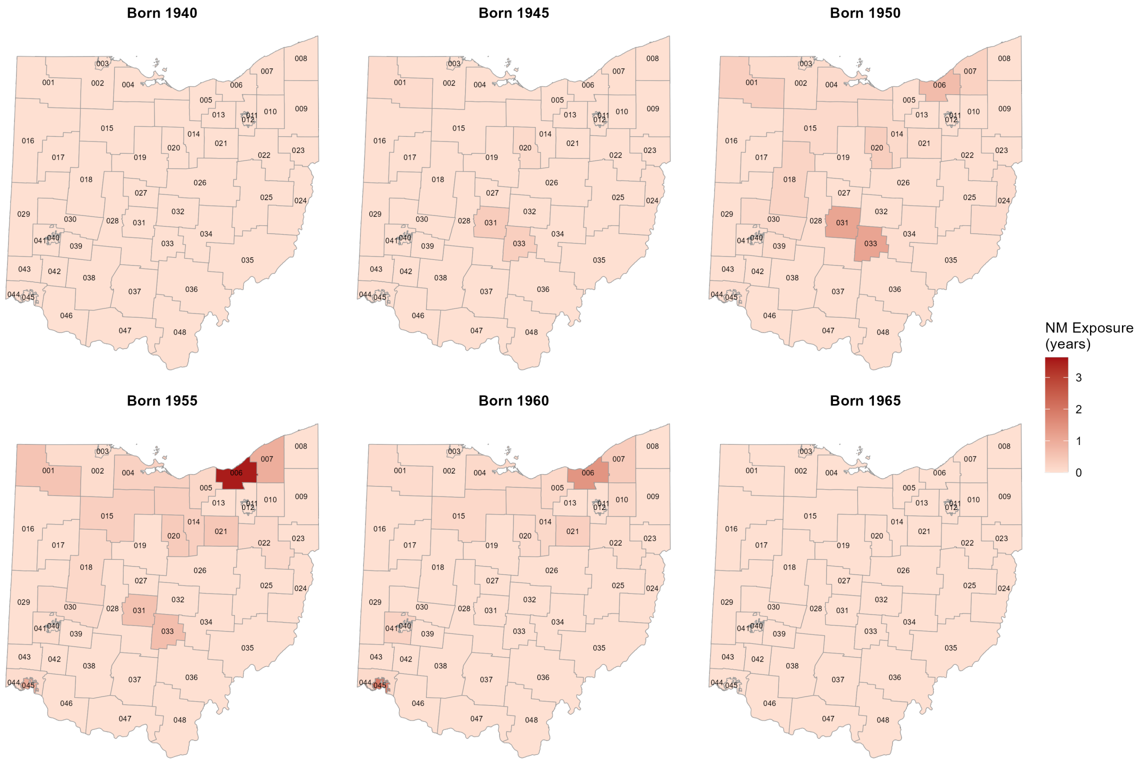
$$Y_{ipct} = \beta_0 + \beta_1 \,\mathrm{NM\ Exposure}_{pcv(t)} + X_i'\delta + \mu_{p\times t} + \mu_c + \varepsilon_{ipct} \tag{4}$$

where Y_{ipct} is either a 1998 O*NET or 1977 DOT task measure, the log of hourly wage, or an indicator equal to one if individual i is employed. NM $\mathrm{Exposure}_{pcv(t)}$ is a continuous measure of expected New Math exposure based on the individual's PUMA of residence and birth cohort which takes a value between 0 and 12.[16] X_i is a vector of individual demographic controls including gender, race, education, and marital status. $\mu_{p\times t}$ and μ_c are PUMA-survey year and cohort fixed effects, respectively. Standard errors are clustered at the 2000 PUMA level, with 1990 PUMAs mapped to their 2000 counterparts to ensure that geographically similar areas are not assigned to different clusters across Censuses.

[15] This assumes populations are distributed evenly across the Census tract, attenuating my result.

[16] I assume individuals were born in the PUMA they currently reside in. This attenuates my result as long as exposure to New Math does not correlate with different propensities to move or if individuals move differentially to areas based on their New Math exposure.

Figure 4: Ohio New Math Exposure by PUMA



Note:

VI Results

Table 2 displays results from estimating Equation 4. Overall, I find that exposure to an abstract math curriculum leads individuals to sort into occupations requiring fewer calculations, as measured by the number facility task, with no corresponding effect on analytic tasks. One additional grade-year of expected New Math exposure decreases an individual's number facility percentile rank by approximately one-fifth of a percentile. This corresponds to approximately half of a percent decrease relative to the mean. I find no effects on other O*NET task measures. I find effects on some DOT task measures, particularly non-routine cognitive analytical, though the DOT does not separately identify abstract and computational math tasks as the O*NET measures do.

My results are consistent with the view that New Math worsened computational skills without producing offsetting gains in abstract mathematical reasoning. This paper suggests this effect to last into adulthood and affect which occupations individuals sort into. Given the numerous data constraints discussed above, my estimates are best interpreted as lower bounds on the true effect of New Math exposure.

Table 1: New Math Task Results

Outcome	(1)	(2)
*Panel A: 1998 O*NET Measures*		
Analytical	-0.319	-0.279
	(0.237)	(0.226)
	{52.536}	{52.555}
Number Facility	-0.432***	-0.394***
	(0.114)	(0.117)
	{52.163}	{52.153}
Routine	0.220	0.161
	(0.145)	(0.139)
	{48.804}	{48.827}
Social Skills	-0.060	0.033
	(0.137)	(0.134)
	{53.768}	{53.726}
Panel B: 1977 DOT Measures		
Non-routine Cognitive Analytical	-0.020	0.043
	(0.169)	(0.184)
	{53.652}	{53.600}
Non-routine Cognitive Interactive	-0.133	-0.041
	(0.122)	(0.199)
	{52.280}	{52.250}
Routine Cognitive	0.480**	0.411**
	(0.219)	(0.293)
	{50.217}	{50.200}
Routine Manual	0.666**	0.582*
	(0.313)	(0.293)
	{48.164}	{48.172}
Non-routine Manual	0.080	0.087
	(0.082)	(0.085)
	{49.917}	{49.895}
Included Cohorts	1945-1962	1940-1962
Gender, Race Controls	X	X
PUMA-Survey Year, Cohort FEs	X	X
N	7,671,475	7,947,670

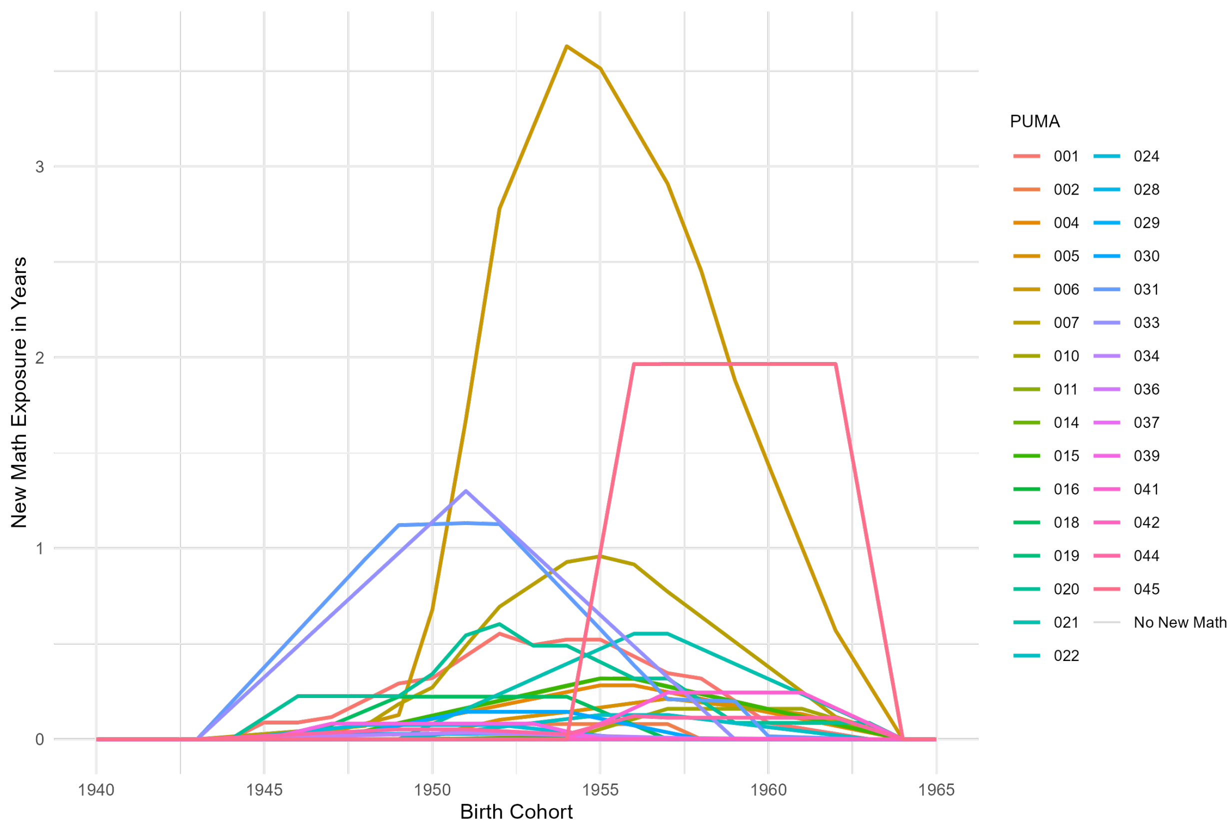
Note: New Math exposure is measured as the population-weighted share of districts, within a PUMA, teaching New Math across the grades a cohort attended, yielding a continuous variable from 0 to 12. Task outcomes, from the 1977 Dictionary of Occupational Titles, are each individuals' occupation-level, employment-weighted percentile rank in the 1980 Census, following Autor, Levy, and Murnane (2003) and Autor and Dorn (2013). N is the number of employed individuals for the included cohorts, who were born in Ohio and only results on employment rate include unemployed individuals. Means on hourly wage regressions are not logged. The estimated coefficient is reported along with its standard error, clustered at the PUMA level, (in parentheses) and outcome average {in braces}. * significant at 10%, ** at 5%, *** at 1%.

Table 2: New Math Employment and Education Results

Outcome	(1)	(2)
Panel A: Labor Market Outcomes		
Log(Hourly Wage)	0.001	0.001
	(0.004)	(0.004)
	{33.186}	{33.211}
$\mathbb{1}$ {Employed}	-0.004	-0.004
	(0.003)	(0.004)
	{0.778}	{0.778}
Panel B: Education Outcomes		
$\mathbb{1}$ {Some High School}	-0.001**	-0.002**
	(0.001)	(0.001)
	{0.983}	{0.982}
$\mathbb{1}$ {High School}	-0.000	0.001
	(0.006)	(0.006)
	{0.414}	{0.412}
$\mathbb{1}$ {Some College}	-0.002	-0.001
	(0.004)	(0.003)
	{0.273}	{0.271}
$\mathbb{1}$ {College}	-0.002	-0.001
	(0.003)	(0.002)
	{0.191}	{0.190}
$\mathbb{1}$ {Graduate School}	-0.001	-0.001
	(0.003)	(0.002)
	{0.065}	{0.065}
Panel C: Other Outcomes		
$\mathbb{1}$ {Married}	-0.005	-0.005
	(0.005)	(0.005)
	{0.652}	{0.655}
Included Cohorts	1945-1962	1940-1962
Gender, Race FEs	X	X
PUMA-Survey Year, Cohort FEs	X	X
N	7,671,475	7,947,670

Note: New Math exposure is measured as the population-weighted share of districts, within a PUMA, teaching New Math across the grades a cohort attended, yielding a continuous variable from 0 to 12. Task outcomes, from the 1977 Dictionary of Occupational Titles, are each individuals' occupation-level, employment-weighted percentile rank in the 1980 Census, following Autor, Levy, and Murnane (2003) and Autor and Dorn (2013). N is the number of employed individuals for the included cohorts, who were born in Ohio and only results on employment rate include unemployed individuals. Means on hourly wage regressions are not logged. The estimated coefficient is reported along with its standard error, clustered at the PUMA level, (in parentheses) and outcome average {in braces}. * significant at 10%, ** at 5%, *** at 1%.

Figure 5: Ohio New Math Exposure by PUMA-Cohort



Note:

VII Appendix

VII.A New Math Curriculum

Table 3: Basic Guidelines of the Greater Cleveland Mathematics Program

Guideline	Description
1	The basic program be suitable for use by all students.
2	The program have a continuous and systematic flow of mathematical concept formation from grades kindergarten through twelve.
3	The program originate at the lowest level of instruction in kindergarten or first grade and be continued through to grade twelve.
4	The teaching approach make the greatest possible use of the discovery method of teaching and provide continuous challenge and stimulation to the student.
5	The program be mathematically correct and pedagogically sound.

Note: Source: ERC 1963, pp. 9.

Table 4: Principles of the Greater Cleveland Mathematics Program

In writing the student and teacher materials, the authors have endeavored to create a program:

Principle	Description
1	That stresses reasoning and understanding.
2	That develops the basic properties of the number system.
3	That leads the child to discover patterns and relations for himself.
4	That uses precise definitions in order to develop the mathematical language.
5	That leads the pupil from the concrete to the abstract.
6	That gives the child an opportunity to discover mathematical principles for himself.
7	That makes use of the child's basic intuitive notions.
8	That takes advantage of the child's natural curiosity and creative imagination.
9	That is self-motivating.
10	That is structured on a continuous flow of ideas.
11	That is suitable for all students.
12	That is comprehensive.

Note: Source: ERC 1963, pp. 10–11.

Table 5: GCMP Sequence, Kindergarten–Grade 3

Concepts	K	1	2	3
Sets				
One-to-one correspondence	✓	✓	✓	✓
Equivalent and non-equivalent sets	✓	✓	✓	✓
Union of sets (addition)	✓	✓	✓	✓
Subsets	✓	✓	✓	✓
Set separation (subtraction)	✓	✓	✓	✓
Sets defining multiplication			✓	✓
Set partitioning (division)			✓	✓
Numbers and Numerals				
Cardinal numbers; numeral distinction	✓	✓	✓	✓
Numbers 1–10	✓	✓	✓	✓
Numbers 0–100		✓	✓	✓
Numbers 0–999			✓	✓
Numbers to millions				✓
Reading/writing numerals	✓	✓	✓	✓
Counting (by 1s; then skip counting)	✓	✓	✓	✓
Place Value				
Base-ten system		✓	✓	✓
Expanded notation		✓	✓	✓
Place value beyond 1,000				✓
Order and Relations				
Ordering and comparing numbers	✓	✓	✓	✓
One-more / one-less	✓	✓	✓	✓
Equations and inequalities		✓	✓	✓
Ordinal numbers	✓	✓	✓	✓
Addition of Whole Numbers				
Meaning of addition; basic facts	✓	✓	✓	✓
Properties of addition	✓	✓	✓	✓
Multi-digit addition (no carrying)		✓	✓	✓
Multi-digit addition (with carrying)			✓	✓
Subtraction of Whole Numbers				
Meaning of subtraction	✓	✓	✓	✓
Inverse relation with addition		✓	✓	✓
Multi-digit subtraction (borrowing)			✓	✓
Multiplication of Whole Numbers				
Repeated addition; set interpretation			✓	✓
Facts through 9×9			✓	✓
Multi-digit multiplication				✓
Division of Whole Numbers				
Partitioning; missing factors			✓	✓
Division facts			✓	✓
Multi-digit division				✓
Fractions				
Parts of a whole; equivalence	✓	✓	✓	✓
Common fractions		✓	✓	✓
Ordering and informal operations				✓
Measurement				
Time and money		✓	✓	✓
Advanced money concepts				✓

Note: From ERC (1963).

VII.B Data

VII.C New Math in Media

Figure 6: Peanuts Comic Strips Referencing New Math (1964–1965)

References

Angrist, J. D. and V. Lavy (1999). “Using Maimonides’ Rule to Estimate the Effect of Class Size on Scholastic Achievement”. *The Quarterly Journal of Economics* 114.2, pp. 533–575.

Autor, D. and D. Dorn (2009). “This Job Is “Getting Old”: Measuring Changes in Job Opportunities Using Occupational Age Structure”. *American Economic Review: Papers & Proceedings* 99.2, pp. 45–51.

Autor, D. and D. Dorn (2013). “The Growth of Low-Skill Service Jobs and the Polarization of the US Labor Market”. *American Economic Review* 103.5, pp. 1553–1597.

Autor, D. and M. J. Handel (2013). “Putting Tasks to the Test: Human Capital, Job Tasks, and Wages”. *Journal of Labor Economics* 31.2, S59–S96.

Autor, D., F. Levy, and R. Murnane (2003). “The Skill Content of Recent Technological Change: An Empirical Exploration”. *The Quarterly Journal of Economics* 118.4, pp. 1279–1333.

Baird, G. H. (Jan. 1961). “The Greater Cleveland Mathematics Program”. *The Mathematics Teacher* 54.1, p. 31.

Baron, E. J. (2022). “School Spending and Student Outcomes: Evidence from Revenue Limit Elections in Wisconsin”. *American Economic Journal: Economic Policy* 14.1, pp. 1–39.

Begle, E. G. (1963). “The SMSG Program”. *The Mathematics Teacher* 56.3, pp. 191–195.

Brickell, H. M. (Dec. 1961). *Organizing New York State for Educational Change. A Study of the Dynamics of Instructional Change in the Elementary and Secondary Schools of New York State with Recommendations for Improved Organization.* Albany, NY: University of the State of New York, State Education Department, p. 127.

Chetty, R., J. N. Friedman, and J. E. Rockoff (2014). “Measuring the Impacts of Teachers I: Evaluating Bias in Teacher Value-Added Estimates”. *American Economic Review* 104.9, pp. 2593–2632.

Deming, D. J. (2017). “The Growing Importance of Social Skills in the Labor Market”. *Quarterly Journal of Economics* 132.4, pp. 1593–1640.

Educational Research Council of Greater Cleveland (1963). *GCMP Greater Cleveland Mathematics Program: Administrator’s Guide.* coordinator L. M. Britton. preliminary edition. Chicago, IL: Science Research Associates, Inc.

Feynman, R. (1965). “New Textbooks for the ’New’ Mathematics”. *Engineering and Science* 28.6, pp. 9–15.

Goldin, C. and L. F. Katz (2008). *The Race between Education and Technology.* Harvard University Press.

Graham, L. (2006). “The Ghost of the Executed Engineer: Technology and the Fall of the Soviet Union”. *Harvard University Press.*

Greater Cleveland Mathematics Program (1962). *Teacher’s Guide for Fourth Grade, Volume 1.* Ed. by P. D. McGarvey. Chicago, IL: Science Research Associates, Inc.

Haeck, C., P. Lefebvre, and P. Merrigan (2014). “The Distributional Impacts of a Universal School Reform on Mathematical Achievements: A Natural Experiment from Canada”. *Economics of Education Review* 41, pp. 137–160.

Hahm, D. W. (2026). “From Curriculum to Career: Early-Career Labor Market Effects of the Common Core”. *Economics of Education Review* 109, p. 1028xx.

Hanushek, E. A. (2003). “The Failure of Input-Based Schooling Policies”. *The Economic Journal* 113.485, F64–F98.

Hanushek, E. A. and S. G. Rivkin (2010). “Generalizations about Using Value-Added Measures of Teacher Quality”. *American Economic Review* 100.2, pp. 267–271.

Hoxby, C. M. (2000). “Does Competition among Public Schools Benefit Students and Taxpayers?” *American Economic Review* 90.5, pp. 1209–1238.

Hunt, T. (2025). “National Defense Education Act”. *Encyclopædia Britannica.*

Kline, M. (1973). “Why Johnny Can’t Add: The Failure of the New Math”. *St. Martin’s Press.*

Kline, M. et al. (1962). “On the Mathematics Curriculum of the High School”. *American Mathematical Monthly* 69.3, pp. 189–193.

Miller, J. W. (Dec. 1990). “Whatever Happened to New Math?” *American Heritage* 41.8, pp. 76–83.

Mueller, F. (Nov. 1966). “The Public Image of “New Mathematics””. *The Mathematics Teacher* 59.7, pp. 618–623.

Nunley, J. M. and R. A. J. Seals (2021). “The Changing Task Content of Jobs for Older Workers in the United States”.

Offner, C. (1982). “Back-to-Basics in Mathematics: An Educational Fraud”. *The Mathematics Teacher* 75.6, pp. 448–452.

Phillips, C. J. (2014a). “In Accordance with a “More Majestic Order”: The New Math and the Nature of Mathematics at Midcentury”. *Isis* 105.3, pp. 540–563.

Phillips, C. J. (2014b). *The New Math: A Political History*. Chicago: University of Chicago Press. ISBN: 9780226184968.

Raimi, R. A. (Jan. 1996). *Whatever Happened to the New Math?* Chronicles Magazine.

Rivkin, S. G., E. A. Hanushek, and J. F. Kain (2005). “Teachers, Schools, and Academic Achievement”. *Econometrica* 73.2, pp. 417–458.

Stanic, G. M. A. and J. Kilpatrick (1992). “Reform in Mathematics Education: The Problem of Implementation”. *American Journal of Education* 100.4, pp. 403–431.

Time (Sept. 22, 1961). “Education: Math Made Interesting”. *Time*.

UICSM Mathematics Project (1970). “UICSM Mathematics Project”. *University of Illinois Archives*, p. 1.